

Sedentary time and cardio-metabolic biomarkers in US adults: NHANES 2003–06



Aims Prolonged sedentary time is ubiquitous in developed economies and is associated with an adverse cardio-metabolic risk profile and premature mortality. This study examined the associations of objectively assessed sedentary time and breaks (interruptions) in sedentary time with continuous cardio-metabolic and inflammatory risk biomarkers, and whether these associations varied by sex, age, and/or race/ethnicity. **Methods and results** Cross-sectional analyses with 4757 participants (≥ 20 years) from the 2003/04 and 2005/06 US National Health and Nutrition Examination Survey (NHANES). An Actigraph accelerometer was used to derive sedentary time [< 100 counts per minute (cpm)] and breaks in sedentary time. Independent of potential confounders, including moderate-to-vigorous exercise, detrimental linear associations (P for trends < 0.05) of sedentary time with waist circumference, HDL-cholesterol, C-reactive protein, triglycerides, insulin, HOMA-%B, and HOMA-%S were observed. Independent of potential confounders and sedentary time, breaks were beneficially associated with waist circumference and C-reactive protein (P for trends < 0.05). There was limited evidence of meaningful differences in associations with biomarkers by age, sex, or race/ethnicity. Notable exceptions were sex-differences in the associations of sedentary time and breaks with HDL-cholesterol, and race/ethnicity differences in the association of sedentary time with waist circumference with associations detrimental in non-Hispanic whites, null in Mexican Americans, and beneficial in non-Hispanic blacks. **Conclusion** These are the first population-representative findings on the deleterious associations of prolonged sedentary time with cardio-metabolic and inflammatory biomarkers. The findings suggest that clinical communications and preventive health messages on reducing and breaking up sedentary time may be beneficial for cardiovascular disease risk.